

Baseline and changes in cardiometabolic index and incident cardiovascular disease in two prospective cohorts

Jiangtao Li^{a,b,c}, Xiang Wei^{a,b,c,*}

^a Division of Cardiovascular Surgery, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

^b Key Laboratory of Organ Transplantation, Ministry of Education, Wuhan, China

^c NHC Key Laboratory of Organ Transplantation, Ministry of Health, Wuhan, China

HIGHLIGHTS

- Participants with upper tertile of baseline cardiometabolic index (CMI) had increased risks of incident cardiovascular disease (CVD) compared to those with lower tertile.
- Compared with tertile 1, participants with tertile 3 of total CMI showed an elevated risk of incident CVD.
- The RCS models showed a nonlinear relationship between baseline and total CMI and incident CVD risks in the CHARLS, while a linear relationship in the ELSA.

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ABSTRACT

Background: The relationship between cardiometabolic index (CMI) and cardiovascular disease (CVD) remains unknown in the general population. Existing research has primarily concentrated on baseline CMI, not taking into consideration the changes in CMI over time.

Objective: This study aimed to investigate the associations of baseline and changes in CMI with the risk of incident CVD.

Methods: This study used data from two prospective cohorts: the China Health and Retirement Longitudinal Study (CHARLS) and the English Longitudinal Study of Ageing (ELSA). Changes in CMI were determined by total CMI, calculated by CMI at baseline plus CMI at the second survey.

Results: A total of 8059 participants from CHARLS and 4279 from ELSA were included in baseline CMI analyses. Participants with upper tertile of baseline CMI had increased risks of incident CVD than those with lower tertile (CHARLS, HR 1.36, 95 % CI 1.21–1.53; ELSA, HR 1.24, 95 % CI 1.07–1.43). A total of 4946 participants from CHARLS and 2371 from ELSA were included in CMI change analyses. For the changes in CMI analyses, participants with tertile 3 showed an elevated risk of incident CVD than tertile 1 (CHARLS: HR 1.53, 95 % CI 1.29–1.81; ELSA: HR 1.41, 95 % CI 1.15–1.73). The restricted cubic spine models showed a nonlinear relationship between baseline and total CMI and incident CVD risks in the CHARLS, while a linear relationship in the ELSA.

Conclusions: Both baseline and total CMI were associated with an increased risk of CVD. Long-term monitoring of CMI levels is of great importance for CVD prevention in clinical practice.

1. Introduction

Cardiovascular disease (CVD) is the leading cause of global mortality and a major contributor to global health loss [1,2]. The overall

prevalence of CVD nearly doubled from 271 million cases in 1990 to 523 million cases in 2019, while the number of deaths due to CVD increased from 12.1 million to 18.6 million globally during the same period [3]. The projected increase in CVD prevalence, particularly among middle-aged and older adults, will place a substantial financial burden on the healthcare system and pose urgent challenges to public health [4].

* Corresponding author at: Division of Cardiovascular Surgery, Tongji Hospital, 1095 Jiefang Ave., Wuhan, 430030, China.

E-mail address: xiangwei@tjh.tjmu.edu.cn (X. Wei).

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These chronic conditions—hypertension, diabetes mellitus, and non-alcoholic fatty liver disease—are major risk factors contributing to the current epidemic of CVD [5–7]. In addition, the CVD burden is further compounded by unhealthy behavioral factors, such as unhealthy diet, smoking, physical inactivity, and obesity [8]. Therefore, the identification of modifiable risk factors is crucial for developing effective prevention strategies and reducing global burden of CVD.

The association between insulin resistance, obesity, and CVD has been well documented [9,10]. The triglyceride/high-density lipoprotein cholesterol (TG/HDL-C) ratio, a good surrogate for insulin resistance, serves as a robust predictor for CVD risk [11,12]. Waist-to-height ratio (WHtR), a combination of waist circumference (WC) and height, can more accurately identify abdominal obesity and evaluate cardiometabolic risk than body mass index [13]. Wakabayashi et al. [14] recently introduced a marker named cardiometabolic index (CMI), calculated as TG/HDL-C ratio multiplied by WHtR, combining insulin resistance and obesity parameters for the effective identification of diabetes. The CMI has been reported to be a more effective predictor of metabolic disorders and provide a comprehensive assessment of the metabolic health, especially regarding the impact on abdominal fat distribution and the protective effect of cholesterol, thus outperforming traditional obesity and lipid indicators [15–18]. Previous studies have found that a higher CMI level is associated with an increased risk of hypertension, kidney disease, and mortality [19–21]. In cohort study of 3752 US adults aged ≥40 years, elevated CMI was significantly associated with all-cause and cardiovascular mortality during an average follow-up of 10.7 years. In addition, a recent study showed a significant association between CMI and the risk of incident CVD in patients with hypertension and obstructive sleep apnea [22]. However, the relationship between CMI and CVD remains unknown in the general population. In addition, prior studies have largely focused on the impact of baseline CMI on study outcomes, not taking into consideration the changes in CMI over time. Recent studies have demonstrated that changes in CMI component, including metabolic parameters and WC, are significantly associated with incident CVD risks [23–26]. Therefore, it is imperative to investigate the associations of changes in CMI with the risk of incident

CVD.

In the current study, we used data from two prospective cohorts: the China Health and Retirement Longitudinal Study (CHARLS) and the English Longitudinal Study of Ageing (ELSA). We aimed to investigate the associations of baseline and changes in CMI with the risk of incident CVD.

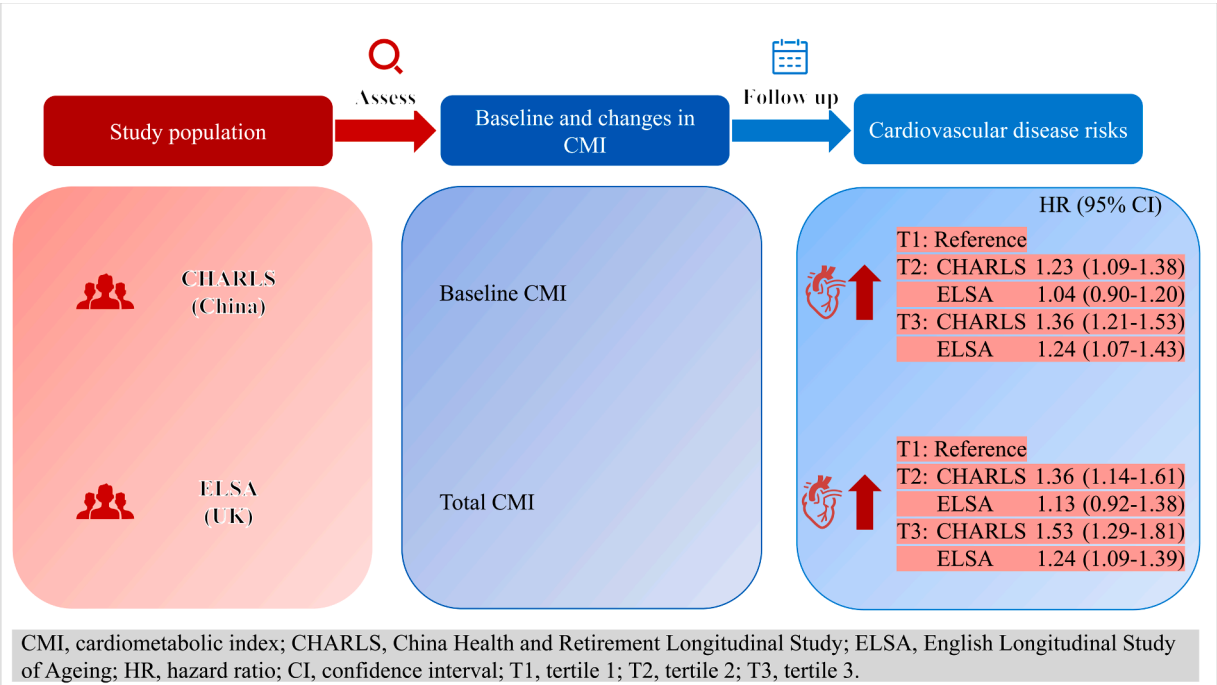
2. Methods

2.1. Study design and data sources

The CHARLS and ELSA are two ongoing prospective and nationally representative cohorts of community-dwelling adults (aged ≥45 years in the CHARLS and ≥50 years in the ELSA), which were conducted in China and the United Kingdom, respectively. Further details regarding the objectives, design, and methods of these two cohorts are available elsewhere [27,28]. The CHARLS received approval from the institutional review board of Peking University, and the ELSA received approval from the Ethics Review Committees of London Multi-Centre Research Ethics. Informed consent was obtained from each participant in these two cohorts and this study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline [29].

In this study, wave 1 (2011) of the CHARLS and wave 2 (2004–2005) of the ELSA were served as the baseline. Wave 3 (2015) of the CHARLS and wave 4 (2008–2009) of the ELSA were served as the second survey. We used data from the baseline and the second survey to evaluate the dynamic change of CMI. Subsequent waves of follow-up were used to track outcomes until the final follow-up wave, which was wave 5 (2020) of the CHARLS and wave 10 (2021–2022) of the ELSA.

The selection process of this study population is shown in Fig. 1. Among 26,473 participants from the CHARLS and ELSA, we excluded 11,415 participants with missing data for calculating the CMI at baseline. Moreover, 2720 participants with CVD at baseline or lost to follow-up were also excluded. Finally, 12,338 eligible participants were included in baseline CMI analyses. For the changes in CMI analyses, we



Central Illustration, Baseline and changes in cardiometabolic index with the risk of incident cardiovascular disease. CMI, cardiometabolic index; CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; HR, hazard ratio; CI, confidence interval; T1, tertile 1; T2, tertile 2; T3, tertile 3.

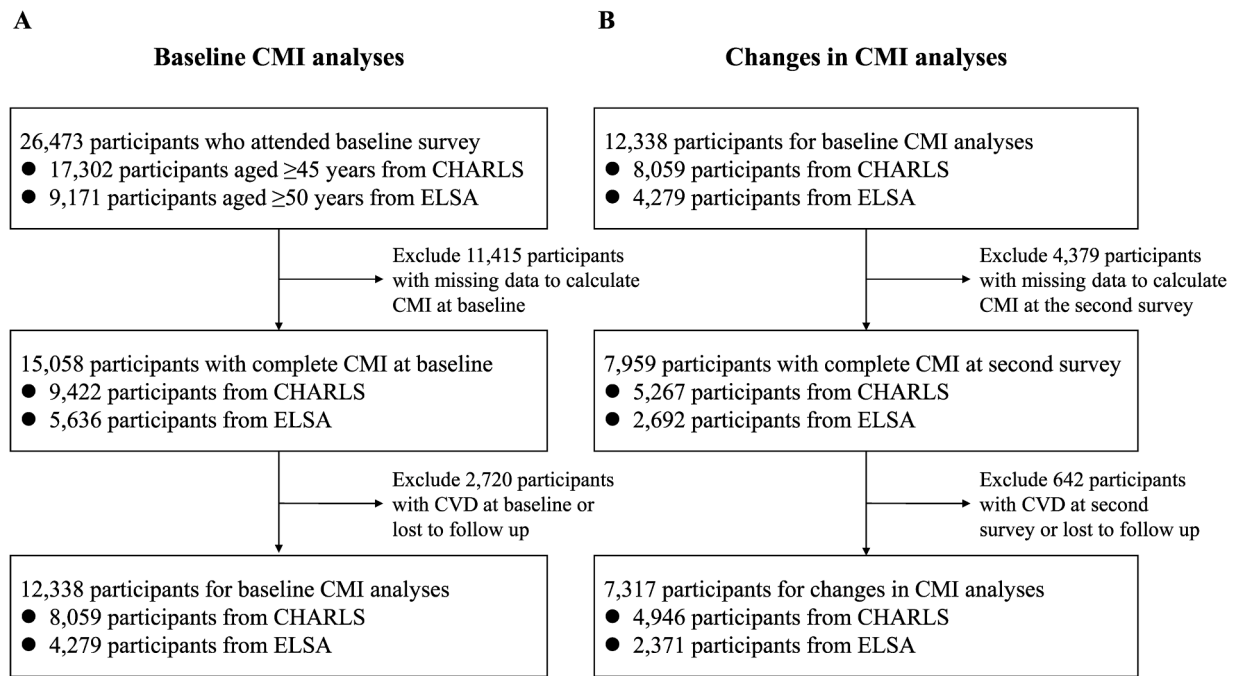


Fig. 1. Selection process of the study population. (A) Participants for baseline CMI analyses were followed from the baseline survey to the last survey. (B) Participants for changes CMI analyses involved in the baseline survey and were followed from the second survey to the last survey. CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; CMI, cardiometabolic index; CVD, cardiovascular disease.

further excluded 5021 participants according to similar criteria. The remaining 7317 participants were included in the final analyses.

2.2. Assessment of cardiometabolic index

The CMI is calculated using the following formula: [14] $\text{CMI} = \text{TG (mmol/L)} / \text{HDL-C (mmol/L)} \times \text{WC (cm)} / \text{height (cm)}$. CMI was treated as a continuous variable and all enrolled participants were stratified into tertiles based on CMI values. In addition, we employed the total CMI to assess the changes in CMI. Total CMI was calculated by the CMI at baseline plus the CMI at the second survey to account for cumulative exposure to cardiometabolic risk. This approach aligns with prior studies demonstrating that longitudinal changes in metabolic and other indices improve CVD prediction [30–33]. By integrating both timepoints, total CMI captures sustained risk beyond a single measurement, mitigating regression dilution bias and reflecting the chronicity of metabolic dysfunction.

2.3. Covariates

The following covariates were included: age, sex, education, marital status, labor force status, smoking status, drinking status, history of hypertension, diabetes, arthritis, chronic lung disease, and cancer, systolic blood pressure (SBP), glycated hemoglobin (HbA1c), low-density lipoprotein cholesterol (LDL-C), and C-reactive protein (CRP). Education was classified into three levels: below high school, high school, and college or above. Marital status was divided into two categories: married or partnered and other marital status (separated, divorced, unmarried, or widowed). Labor force status was divided into two categories: working and not working. Smoking status was categorized as current smokers and former/never smokers. Similarly, drinking status was categorized as current drinkers and former/never drinkers.

2.4. Definitions of outcomes

The primary outcome of this study was incident CVD. Consistent with previous studies, [25,30] CVD was ascertained based on

self-reported physician-diagnosed heart disease or stroke. Participants were asked in each wave, "Have you been told by a doctor that you have been diagnosed with a heart disease, including angina, heart attack, congestive heart failure, and other heart problems?" and "Have you been told by a doctor that you have been diagnosed with a stroke?" Those who reported being diagnosed with heart disease or stroke were considered to have CVD. In subsequent waves, participants were required to confirm the existence of heart disease and stroke if they reported those in the last wave. If participants disputed self-reported heart disease or stroke from previous waves, they were corrected retrospectively.

In the baseline CMI analyses, follow-up was initiated from wave 1 (2012) of the CHARLS and wave 2 (2004–2005) of the ELSA. In the changes in CMI analyses, follow-up was initiated from wave 3 (2015) of the CHARLS and wave 4 (2008–2009) of the ELSA. The endpoint of follow-up was at the first occurrence of CVD, death, or the censoring date, whichever came first. The censoring date was defined as the date of the last survey attended by each participant. Ideally, the last survey was wave 5 (2020) of the CHARLS and wave 10 (2020–2021) of the ELSA. Death data were available up to the last survey in the CHARLS, while those data were only available up to wave 6 (2012–2013) in the ELSA [34].

2.5. Statistical analyses

The idea of this study was conceived in April 2024, and the final data acquisition and analysis were conducted in December 2024. For descriptive statistics, continuous variables were expressed as mean and standard deviation (SD) or median and interquartile range (IQR), while categorical variables were expressed as numbers (percentages). The missing rates of covariates were detailed in Supplemental Tables 1 and 2. Missing data of covariates were imputed using multiple imputations with chained equations [35].

To analyze the association of baseline CMI with the risk of incident CVD, we utilized the Cox proportional hazards regression to calculate the hazard ratio (HR) and its 95 % confidence interval (CI). Baseline CMI was classified into three groups. T1 was the lower tertile, T2 was the middle tertile, and T3 was the upper tertile. In the CHARLS, T1 of

baseline CMI < 0.349, 0.349 ≤ T2 of baseline CMI < 0.689, T3 of baseline CMI ≥ 0.689. In the ELSA, T1 of baseline CMI < 0.417, 0.417 ≤ T2 of baseline CMI < 0.792, T3 of baseline CMI ≥ 0.792. Two models were constructed for the Cox regression using participants in the lower tertile of baseline CMI as the reference. Model 1 was unadjusted. Model 2 was adjusted for age, sex, education, marital status, labor force status, smoking status, drinking status, history of hypertension, diabetes, arthritis, chronic lung disease, and cancer, SBP, HbA1c, LDL-C, and CRP. We confirmed the proportional hazards assumption for these models using the Schoenfeld residual. The trend test was conducted using the median value of each tertile of baseline CMI. In addition, we fitted restricted cubic spline (RCS) regression models to determine the presence of a nonlinear relationship between baseline CMI and incident CVD, with three knots positioned at the 10th, 50th, and 90th percentiles. Using similar methods, we also analyzed associations of changes in CMI with incident CVD. Total CMI was also divided into three groups. In the CHARLS, T1 of total CMI < 0.817, 0.817 ≤ T2 of total CMI < 1.521, T3 of total CMI ≥ 1.521. In the ELSA, T1 of total CMI < 0.865, 0.865 ≤ T2 of total CMI < 1.524, T3 of total CMI ≥ 1.524. For change in CMI analyses, we further controlled for the baseline CMI in the multivariable-adjusted model.

Several sensitivity analyses were performed: (1) to reduce the potential reverse causation bias, the main analyses were repeated after excluding participants who developed CVD in the first two years of follow-up; (2) we repeated analyses of changes in CMI after further adjusting for the baseline CMI; (3) considering the competing risk between mortality and incident heart disease, the main analyses were repeated using the competing risk model; [36] (4) we repeated the main analyses after further adjusting for the use of antihypertensive drugs and antidiabetic drugs. In addition, the main analyses were repeated when using the outcomes using the outcomes of heart disease and stroke separately. Finally, we conducted stratified analyses by age (middle-aged: <65 years; older: ≥65 years) and sex (male and female).

All statistical analyses were performed using R software (version 4.3.2). A two-tailed *P* value < 0.05 was considered statistically significant.

3. Results

3.1. Baseline characteristics of the study population

Based on the inclusion and exclusion criteria, 8059 participants from the CHARLS (mean age: 59.2 years, male: 47.4 %) and 4279 from the ELSA (mean age: 64.7 years, male: 44.2 %) were included in the baseline CMI analyses. Baseline characteristics of these participants with different CMI tertiles are shown in Table 1. Participants with higher tertiles of CMI were more likely to be working and tended to have a history of hypertension and diabetes than those with the lowest tertile of CMI. Participants with higher tertiles of CMI also exhibited higher levels of SBP, HbA1c, and CRP in comparison to those with the lowest tertile of CMI. In the analyses of changes in CMI, 4946 participants from the CHARLS (mean age: 58.6 years, male: 46.4 %) and 2371 from the ELSA (mean age: 63.1 years, male: 42.3 %) were included according to corresponding criteria. Baseline characteristics of these participants are shown in Table 2. In addition, we also presented the baseline characteristics of these participants using data not being imputed (Supplemental Tables 1 and 2). The results were consistent with those in Tables 1 and 2.

3.2. Association of baseline cardiometabolic index with incident cardiovascular disease

In the baseline CMI analyses, the median follow-up periods were 9.0 years in the CHARLS and 11.7 years in the ELSA. During the follow-up, 1845 incident CVD events (1395 heart disease and 648 strokes) in the CHARLS and 1163 incident CVD events (1030 heart disease and 266

Table 1
Baseline characteristics of participants for baseline cardiometabolic index analyses.

Variables	Baseline CMI of CHARLS (n = 8059)			Baseline CMI of ELSA (n = 4279)		
	T1	T2	T3	T1	T2	T3
Number (%)	2686 (33.3)	2686 (33.3)	2687 (33.3)	1426 (33.3)	1426 (33.3)	1427 (33.3)
Age, mean (SD), years	59.6 (9.7)	59.3 (9.4)	58.6 (9.0)	64.2 (9.2)	65.4 (9.3)	64.4 (8.8)
Sex, n (%)						
Male	1433 (53.4)	1237 (46.1)	1153 (42.9)	462.0 (32.4)	620.0 (43.5)	810.0 (56.8)
Female	1253 (46.6)	1449 (53.9)	1534 (57.1)	964.0 (67.6)	806.0 (56.5)	617.0 (43.2)
Education, n (%)						
Below high school	2442 (90.9)	2423 (90.2)	2415 (89.9)	580.0 (40.7)	658.0 (46.1)	727.0 (50.9)
High school	210 (7.8)	230 (8.6)	241 (9.0)	409.0 (28.7)	387.0 (27.1)	338.0 (23.7)
College or above	34 (1.3)	33 (1.2)	31 (1.2)	437.0 (30.6)	381.0 (26.7)	362.0 (25.4)
Marital status, n (%)						
Married or partnered	2347 (87.4)	2356 (87.7)	2387 (88.8)	955.0 (67.0)	999.0 (70.1)	1016.0 (71.2)
Other marital status	339 (12.6)	330 (12.3)	300 (11.2)	471.0 (33.0)	427.0 (29.9)	411.0 (28.8)
Labor force status, n (%)						
Working	1862 (69.3)	1657 (61.7)	1535 (57.1)	557.0 (39.1)	492.0 (34.5)	547.0 (38.3)
Not working	824 (30.7)	1029 (38.3)	1152 (42.9)	869.0 (60.9)	934.0 (65.5)	880.0 (61.7)
Smoking status, n (%)						
Current smokers	1178 (43.9)	1034 (38.5)	975 (36.3)	806.0 (56.5)	869.0 (60.9)	935.0 (65.5)
Former/never smokers	1508 (56.1)	1652 (61.5)	1712 (63.7)	620.0 (43.5)	557.0 (39.1)	492.0 (34.5)
Drinking status, n (%)						
Current drinkers	1274 (47.4)	1086 (40.4)	1031 (38.4)	1332.0 (93.4)	1337.0 (93.8)	1289.0 (90.3)
Former/never drinkers	1412 (52.6)	1600 (59.6)	1656 (61.6)	94.0 (6.6)	89.0 (6.2)	138.0 (9.7)
Hypertension, n (%)						
Yes	785 (29.2)	970 (36.1)	1287 (47.9)	659.0 (46.2)	834.0 (58.5)	906.0 (63.5)
No	1901 (70.8)	1716 (63.9)	1400 (52.1)	767.0 (53.8)	592.0 (41.5)	521.0 (36.5)
Diabetes, n (%)						
Yes	231 (8.6)	320 (11.9)	583 (21.7)	41.0 (2.9)	97.0 (6.8)	196.0 (13.7)
No	2455 (91.4)	2366 (88.1)	2104 (78.3)	1385.0 (97.1)	1329.0 (93.2)	1231.0 (86.3)
Arthritis, n (%)						
Yes	878 (32.7)	949 (35.3)	966 (36.0)	448.0 (31.4)	482.0 (33.8)	523.0 (36.7)
No	1808 (67.3)	1737 (64.7)	1721 (64.0)	978.0 (68.6)	944.0 (66.2)	904.0 (63.3)
Chronic lung disease, n (%)						
Yes	290 (10.8)	253 (9.4)	222 (8.3)	78.0 (5.5)	86.0 (6.0)	86.0 (6.0)
No	2396 (89.2)	2433 (90.6)	2465 (91.7)	1348.0 (94.5)	1340.0 (94.0)	1341.0 (94.0)
Cancer, n (%)						
Yes	23 (0.9)	29 (1.1)	26 (1.0)	96.0 (6.7)	90.0 (6.3)	104.0 (7.3)
No	2663 (99.1)	2657 (98.9)	2661 (99.0)	1330.0 (93.3)	1336.0 (93.7)	1323.0 (92.7)

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Table 1 (continued)

Variables	Baseline CMI of CHARLS (n = 8059)			Baseline CMI of ELSA (n = 4279)		
	T1	T2	T3	T1	T2	T3
SBP, mean (SD), mmHg	125.4 (20.6)	128.7 (20.9)	132.6 (21.3)	132.6 (18.8)	136.8 (18.2)	137.5 (17.3)
HbA1c, mean (SD), %	5.2 (0.6)	5.3 (0.7)	5.4 (1.0)	5.4 (0.4)	5.5 (0.5)	5.7 (0.8)
LDL-C, mean (SD), mmol/L	2.9 (0.8)	3.1 (0.9)	3.0 (1.0)	3.7 (0.9)	3.9 (1.0)	3.6 (1.0)
CRP, median (IQR), mg/L	0.8 (0.5, 1.7)	1.0 (0.5, 2.1)	1.3 (0.7, 2.6)	1.2 (0.6, 2.5)	2.1 (1.1, 4.3)	2.6 (1.3, 5.2)

CMI, cardiometabolic index; CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; SBP, systolic blood pressure; HbA1c, glycated hemoglobin; LDL-C, low-density lipoprotein cholesterol; CRP, C-reactive protein.

strokes) in the ELSA were recorded. Table 3 displays the association between the baseline CMI with the risk of incident CVD. Participants with the upper tertile of baseline CMI presented significantly increased risks of incident CVD compared to those with the lower tertile (CHARLS, HR 1.36, 95 % CI 1.21–1.53; ELSA, HR 1.24, 95 % CI 1.07–1.43). For participants with the middle tertile of baseline CMI, significantly increased CVD risks were only observed in the CHARLS (HR 1.23, 95 % CI 1.09–1.38), but not in the ELSA. Moreover, the trend tests revealed increasing trends of incident CVD risks with increasing baseline CMI in the CHARLS and ELSA (all P for trend <0.001). In the multivariable-adjusted RCS regression models, a linear association was observed between the baseline CMI and the risk of incident CVD in the ELSA (P for overall = 0.009; P for nonlinear = 0.765). However, a nonlinear relationship was observed in the CHARLS (P for overall <0.001; P for nonlinear = 0.005) and the truncated CMI value is 0.88, representing the 76th percentile of the baseline CMI (Fig. 2).

3.3. Association of total cardiometabolic index with incident cardiovascular disease

In the changes in CMI analyses, the median follow-up periods were 5.0 and 10.1 years in the CHARLS and ELSA, respectively. During the follow-up, 897 incident CVD events (603 heart disease and 380 strokes) in the CHARLS and 564 incident CVD events (491 heart disease and 111 strokes) in the ELSA were recorded. Table 4 displays the association between total CMI with the risk of incident CVD. Participants with the upper tertile of total CMI had significantly increased risks of incident CVD compared to those with the lower tertile (CHARLS, HR 1.53, 95 % CI 1.29–1.81; ELSA, HR 1.41, 95 % CI 1.15–1.73). For participants with the middle tertile of total CMI, significantly increased CVD risks were only observed in the CHARLS (HR 1.36, 95 % CI 1.14–1.61), but not in the ELSA. Moreover, the trend tests indicated that there were increasing trends of incident CVD risks with the increasing total CMI in the CHARLS and ELSA (all P for trend <0.001). In the multivariable-adjusted RCS regression models, there was a linear association was observed between the total CMI and the risk of incident CVD in the ELSA (P for overall = 0.002; P for nonlinear = 0.534), while a positive nonlinear relationship was observed in the CHARLS (P for overall <0.001; P for nonlinear = 0.017) (Fig. 3).

3.4. Sensitivity analyses

After excluding participants who developed CVD in the first two years of follow-up, consistent results were observed (Supplemental Tables 5 and 6). Participants with higher tertiles of both baseline and total CMI still had increased risks of incident CVD compared to those with lower tertiles. The association between total CMI and incident CVD risks

Table 2

Baseline characteristics of participants for changes in cardiometabolic index analyses.

Variables	Total CMI of CHARLS (n = 4946)			Total CMI of ELSA (n = 2371)		
	T1	T2	T3	T1	T2	T3
Number (%)	1649 (33.3)	1648 (33.3)	1649 (33.3)	790 (33.3)	790 (33.3)	791 (33.3)
Age, mean (SD), years	59.1 (8.9)	59.0 (8.9)	57.8 (8.3)	62.7 (7.8)	63.8 (8.5)	62.8 (7.7)
Sex, n (%)						
Male	896 (54.3)	707 (42.9)	693 (42.0)	252.0 (31.9)	318.0 (40.3)	433.0 (54.7)
Female	753 (45.7)	941 (57.1)	956 (58.0)	538.0 (68.1)	472.0 (59.7)	358.0 (45.3)
Education, n (%)						
Below high school	1514 (91.8)	1502 (91.1)	1497 (90.8)	264.0 (33.4)	339.0 (42.9)	360.0 (45.5)
High school	117 (7.1)	128 (7.8)	136 (8.2)	261.0 (33.0)	206.0 (26.1)	212.0 (26.8)
College or above	18 (1.1)	18 (1.1)	16 (1.0)	265.0 (33.5)	245.0 (31.0)	219.0 (27.7)
Marital status, n (%)						
Married or partnered	1477 (89.6)	1451 (88.0)	1500 (91.0)	551.0 (69.7)	565.0 (71.5)	585.0 (74.0)
Other marital status	172 (10.4)	197 (12.0)	149 (9.0)	239.0 (30.3)	225.0 (28.5)	206.0 (26.0)
Labor force status, n (%)						
Working	1221 (74.0)	1071 (65.0)	1016 (61.6)	351.0 (44.4)	319.0 (40.4)	330.0 (41.7)
Not working	428 (26.0)	577 (35.0)	633 (38.4)	439.0 (55.6)	471.0 (59.6)	461.0 (58.3)
Smoking status, n (%)						
Current smokers	744 (45.1)	570 (34.6)	590 (35.8)	429.0 (54.3)	462.0 (58.5)	512.0 (64.7)
Former/never smokers	905 (54.9)	1078 (65.4)	1059 (64.2)	361.0 (45.7)	328.0 (41.5)	279.0 (35.3)
Drinking status, n (%)						
Current drinkers	789 (47.8)	614 (37.3)	636 (38.6)	755.0 (95.6)	742.0 (93.9)	735.0 (92.9)
Former/never drinkers	860 (52.2)	1034 (62.7)	1013 (61.4)	35.0 (4.4)	48.0 (6.1)	56.0 (7.1)
Hypertension, n (%)						
Yes	439 (26.6)	586 (35.6)	770 (46.7)	332.0 (42.0)	401.0 (50.8)	479.0 (60.6)
No	1210 (73.4)	1062 (64.4)	879 (53.3)	458.0 (58.0)	389.0 (49.2)	312.0 (39.4)
Diabetes, n (%)						
Yes	148 (9.0)	181 (11.0)	333 (20.2)	23.0 (2.9)	49.0 (6.2)	89.0 (11.3)
No	1501 (91.0)	1467 (89.0)	1316 (79.8)	767.0 (97.1)	741.0 (93.8)	702.0 (88.7)
Arthritis, n (%)						
Yes	554 (33.6)	567 (34.4)	556 (33.7)	246.0 (31.1)	266.0 (33.7)	268.0 (33.9)
No	1095 (66.4)	1081 (65.6)	1093 (66.3)	544.0 (68.9)	524.0 (66.3)	523.0 (66.1)
Chronic lung disease, n (%)						
Yes	166 (10.1)	143 (8.7)	133 (8.1)	35.0 (4.4)	30.0 (3.8)	38.0 (4.8)
No	1483 (89.9)	1505 (91.3)	1516 (91.9)	755.0 (95.6)	760.0 (96.2)	753.0 (95.2)
Cancer, n (%)						
Yes	13 (0.8)	16 (1.0)	18 (1.1)	45.0 (5.7)	50.0 (6.3)	45.0 (5.7)
No	1636 (99.2)	1632 (99.0)	1631 (98.9)	745.0 (94.3)	740.0 (93.7)	746.0 (94.3)
SBP, mean (SD), mmHg	124.1 (19.9)	128.1 (20.1)	131.6 (21.0)	131.4 (18.2)	134.4 (17.2)	136.7 (17.3)

(continued on next page)

Table 2 (continued)

Variables	Total CMI of CHARLS (n = 4946)			Total CMI of ELSA (n = 2371)		
	T1	T2	T3	T1	T2	T3
HbA1c, mean (SD), %	5.1 (0.6)	5.2 (0.7)	5.4 (1.0)	5.4 (0.4)	5.5 (0.5)	5.6 (0.8)
LDL-C, mean (SD), mmol/L	2.9 (0.8)	3.1 (0.9)	3.0 (1.0)	3.7 (0.9)	3.9 (1.0)	3.7 (1.0)
CRP, median (IQR), mg/L	0.7 (0.4, 1.5)	1.0 (0.5, 2.0)	1.3 (0.7, 2.4)	1.1 (0.6, 2.4)	2.0 (1.0, 3.8)	2.4 (1.3, 4.4)

CMI, cardiometabolic index; CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; SBP, systolic blood pressure; HbA1c, glycated hemoglobin; LDL-C, low-density lipoprotein cholesterol; CRP, C-reactive protein.

were remained consistent after further adjusting for the baseline CMI (Supplemental Table 7). Furthermore, the results were consistent with the main analyses after further conducting the competing risk analyses between mortality and CVD (Supplemental Tables 8 and 9) or adjusting for the uses of antihypertensive and antidiabetic drugs (Supplemental Tables 10 and 11). The association of CMI with heart disease and stroke were also consistent with CVD (Supplemental Tables 12 and 13). In the stratified analyses, similar results were observed in middle-aged and older participants (Supplemental Tables 14 and 15). Meanwhile, the results were similar in males and females (Supplemental Tables 16 and

17).

4. Discussion

In this study with two prospective cohorts, we examined the associations of baseline and changes in CMI with the risk of incident CVD (Graphical Abstract). Higher levels of the baseline CMI were associated with an increased risk of incident CVD compared to lower levels. Moreover, participants with higher levels of total CMI had elevated risks of incident CVD. The dose-response curve between the baseline and total CMI and CVD risk showed a nonlinear relationship in the CHARLS, while a linear relationship in the ELSA. Therefore, monitoring CMI offers a simple yet potent approach to CVD prevention strategies in clinical practice.

CMI, a newly developed clinical marker, was first reported by Wakabayashi et al. in 2015 and has been shown to have a significant advantage in identifying diabetes [14] In a cohort study of 7347 participants from China, participants with high levels of CMI had an increased risk of incident diabetes than those with low levels (HR 1.78, 95 %CI 1.55–2.05) [37] Moreover, recent studies have confirmed CMI's linkage to increased risks of new-onset diabetes and expanded its relevance to incident CVD [22,38] During a median follow-up of 6.8 years, Cai et al. [22] in a retrospective cohort study enrolling 2067 participants found a positive association between CMI and the risk of incident CVD in patients with hypertension and obstructive sleep apnea (per standard deviation increment, HR 1.31, 95 % CI 1.20–1.43). Consistently, our

Table 3
Association of baseline cardiometabolic index with the risk of incident cardiovascular disease.

	CHARLS (n = 8059)					ELSA (n = 4279)				
	Case, n (%)	HR1 (95 % CI) ^a	PI ^a	HR2 (95 % CI) ^b	P2 ^b	Case, n (%)	HR1 (95 % CI) ^a	PI ^a	HR2 (95 % CI) ^b	P2 ^b
T1 of baseline CMI	508 (18.9)	1 (reference)		1 (reference)		346 (24.3)	1 (reference)		1 (reference)	
T2 of baseline CMI	635 (23.6)	1.29 (1.15–1.45)	<0.001	1.23 (1.09–1.38)	<0.001	381 (26.7)	1.15 (1.00–1.34)	0.053	1.04 (0.90–1.20)	0.617
T3 of baseline CMI	702 (26.1)	1.44 (1.29–1.62)	<0.001	1.36 (1.21–1.53)	<0.001	436 (30.6)	1.41 (1.22–1.62)	<0.001	1.24 (1.07–1.43)	0.005
P for trend ^c		1.43 (1.27–1.62)	<0.001	1.35 (1.19–1.53)	<0.001		1.45 (1.25–1.69)	<0.001	1.28 (1.09–1.51)	0.003

T1 was the lower tertile, T2 was the middle tertile, and T3 was the upper tertile. In the CHARLS, T1 of baseline CMI < 0.349, 0.349 ≤ T2 of baseline CMI < 0.689, T3 of baseline CMI ≥ 0.689. In the ELSA, T1 of baseline CMI < 0.417, 0.417 ≤ T2 of baseline CMI < 0.792, T3 of baseline CMI ≥ 0.792.

CMI, cardiometabolic index; CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; HR, hazard ratio; CI, confidence interval.

^a HR and P were unadjusted.
^b HR and P were adjusted for age, sex, education, marital status, labor force status, smoking status, drinking status, history of hypertension, diabetes, arthritis, chronic lung disease, and cancer, systolic blood pressure, glycated hemoglobin, low-density lipoprotein cholesterol, and C-reactive protein.
^c Tests for linear trends across four categories of baseline CMI were performed by modeling the median value within each category as a continuous variable.

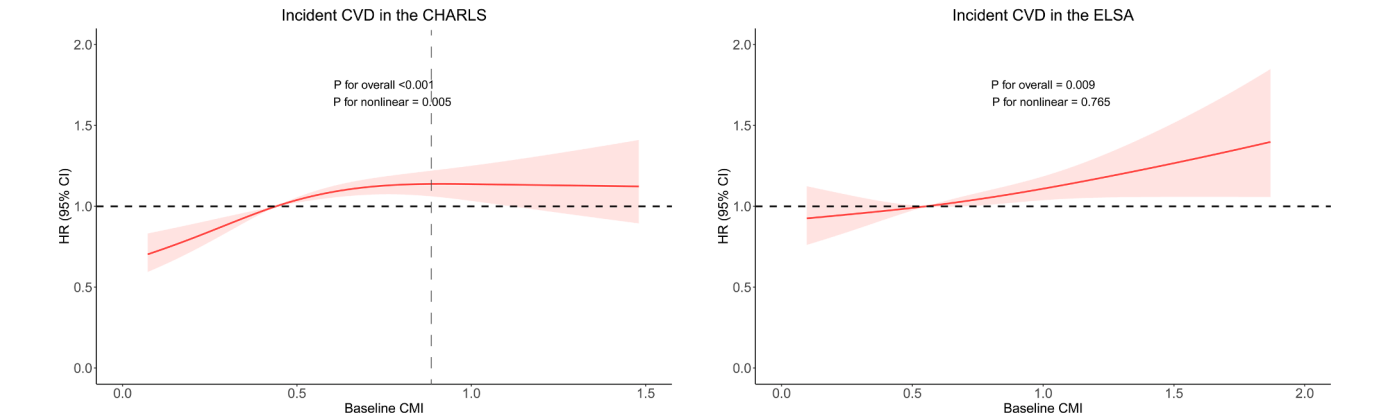


Fig. 2. Restricted cubic spline curves for incident CVD according to the baseline CMI. HRs and 95 % CIs derived from restricted cubic spline regression, with knots placed at the 10th, 50th, and 90th percentiles of the distribution of baseline CMI. The horizontal dotted line represents the HR of 1.0. HRs were adjusted for age, sex, education, marital status, labor force status, smoking status, drinking status, history of hypertension, diabetes, arthritis, chronic lung disease, and cancer, systolic blood pressure, glycated hemoglobin, low-density lipoprotein cholesterol, and C-reactive protein. CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; CMI, cardiometabolic index; CVD, cardiovascular disease; HR, hazard ratio; CI, confidence interval.

Table 4
Association of total cardiometabolic index with the risk of incident cardiovascular disease.

	CHARLS (n = 4946)					ELSA (n = 2371)				
	Case, n (%)	HR1 (95 % CI) ^a	P1 ^a	HR2 (95 % CI) ^b	P2 ^b	Case, n (%)	HR1 (95 % CI) ^a	P1 ^a	HR2 (95 % CI) ^b	P2 ^b
T1 of total CMI	230 (13.9)	1 (reference)		1 (reference)		161 (20.4)	1 (reference)		1 (reference)	
T2 of total CMI	316 (19.2)	1.43 (1.20–1.69)	<0.001	1.36 (1.14–1.61)	<0.001	189 (23.9)	1.22 (1.00–1.49)	0.053	1.13 (0.92–1.38)	0.250
T3 of total CMI	351 (21.3)	1.62 (1.37–1.91)	<0.001	1.53 (1.29–1.81)	<0.001	214 (27.1)	1.53 (1.26–1.86)	<0.001	1.41 (1.15–1.73)	<0.001
P for trend ^c		1.27 (1.16–1.39)	<0.001	1.23 (1.12–1.35)	<0.001		1.29 (1.15–1.44)	<0.001	1.24 (1.09–1.39)	<0.001

Total CMI was calculated by the CMI at baseline plus the CMI at the second survey. T1 was the lower tertile, T2 was the middle tertile, and T3 was the upper tertile. In the CHARLS, T1 of total CMI < 0.817, 0.817 ≤ T2 of total CMI < 1.521, T3 of total CMI ≥ 1.521. In the ELSA, T1 of total CMI < 0.865, 0.865 ≤ T2 of total CMI < 1.524, T3 of total CMI ≥ 1.524.

CMI, cardiometabolic index; CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; HR, hazard ratio; CI, confidence interval.

- ^a HR and P were unadjusted.
^b HR and P were adjusted for age, sex, education, marital status, labor force status, smoking status, drinking status, history of hypertension, diabetes, arthritis, chronic lung disease, and cancer, systolic blood pressure, glycated hemoglobin, low-density lipoprotein cholesterol, and C-reactive protein.
^c Tests for linear trends across four categories of total CMI were performed by modeling the median value within each category as a continuous variable.

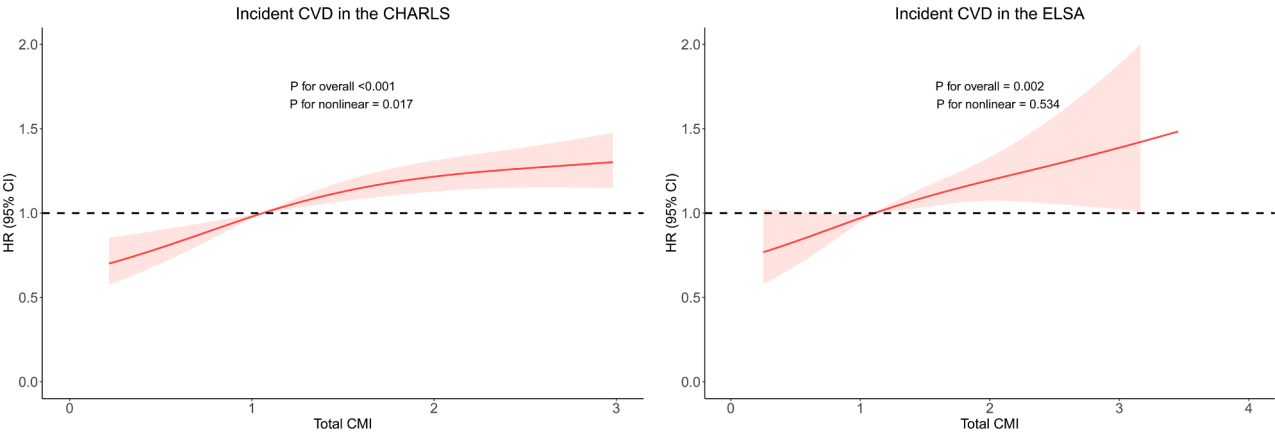


Fig. 3. Restricted cubic spline curves for incident CVD according to the total CMI. Total CMI was calculated by the CMI at baseline plus the CMI at the second survey. HRs and 95 % CIs derived from restricted cubic spline regression, with knots placed at the 10th, 50th, and 90th percentiles of the distribution of baseline CMI. The horizontal dotted line represents the HR of 1.0. HRs were adjusted for age, sex, education, marital status, labor force status, smoking status, drinking status, history of hypertension, diabetes, arthritis, chronic lung disease, and cancer, systolic blood pressure, glycated hemoglobin, low-density lipoprotein cholesterol, and C-reactive protein. CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; CMI, cardiometabolic index; CVD, cardiovascular disease; HR, hazard ratio; CI, confidence interval.

study found elevated risks of incident CVD in participants with higher levels of CMI compared to those with lower levels. Recent research suggests that biological mechanisms linking CMI with CVD might be shared physiological pathways, including the release of pro-inflammatory cytokines and chronic inflammation, which have been elucidated as crucial contributors to atherosclerosis, resulting in the progress of CVD [39–41]. Nevertheless, our results were still significant after adjusting for traditional risk factors of CVD. Our findings suggested that elevated CMI was associated with incident CVD in these two cohorts, though further studies are needed to confirm causality.

In addition to the baseline CMI, our study additionally investigated the associations of changes in CMI with incident CVD, which were not examined previously. Both WHtR and TG/HDL-C ratio, which are utilized to calculate the CMI, are considered time-varying exposures. However, the above evidence [22,38] is susceptible to potential regression dilution bias due to a single measurement of the CMI [42]. It has been suggested that the above limitations and methodological insufficiencies can be addressed by assessing the impact of changes in CMI on incident CVD risks. Total CMI, calculated by the CMI at baseline plus the CMI at the second survey, was used to assess the changes in CMI. In this study, we found that total CMI were associated with an increased risk of incident CVD. The highest risk of incident CVD was observed in participants with the highest tertile of total CMI, with multivariate-adjusted HR of 1.53 (95 % CI, 1.29–1.81) in the CHARLS

and HR of 1.41 (95 % CI, 1.15–1.73) in the ELSA. In addition, the risks were not attenuated by several sensitivity analyses, which further corroborated our previous findings.

Our study has important clinical and public health implications. First, this study contributes to the existing knowledge of CVD risk prevention and control by providing evidence supporting the utilization of the baseline and total CMI as a clinically valuable marker for identifying individuals at high risk of developing CVD. The CMI, calculated by TG, HDL-C, WC, and height, can be conveniently obtained from a single sample at the same time, even in primary healthcare settings, enhancing the generalizability and practicality of CMI in both clinical and epidemiological studies. For the general population, it is important to maintain appropriate levels of TG, HDL-C, and WC within the desirable range for preventing chronic diseases. In addition, for those with hyperlipidemia and obesity, it is vital to encourage adhere to lipid-lowering and weight-lowering drugs for actively controlling CMI levels. Therefore, integrating the assessment of CMI into routine cardiovascular practice is necessary. Furthermore, through dynamic monitoring of CMI, we can not only assess the CVD risk of participants at any time but also adjust the clinical interventions according to the changes in CMI at any time, enabling precise medical treatment and reducing the CVD risk and the burden of CVD.

This study has several strengths. To the best of our knowledge, this is the first study to assess the associations between changes in CMI and the

risk of incident CVD. Furthermore, this study included two prospective cohorts from different ethnicities, featuring rigorous study designs, large sample sizes, and long-term follow-up. In addition, the diverse sensitivity analyses ensured the robustness of our findings.

There are some limitations in this study. First, the determination of CVD relied on self-reported physician-diagnosed information, which might lead to a misclassification bias. However, a previous study [43] found that 77.5 % of self-reported coronary heart disease were consistent with the medical records in the ELSA, which suggested that the potential misclassification bias was minor. In addition, we performed sensitivity analyses using the outcomes of heart disease and stroke separately. These results were consistent with the main analyses. Second, selection bias might occur for changes in CMI analyses as we further excluded 5021 participants from the baseline CMI analyses. We compared the baseline characteristics of included and excluded participants (Supplemental Tables 18 and 19), indicating the existence of healthy participant bias. Third, there was no information on statin therapy in both databases, which had an impact on the lipid levels. Thus, it is not clear whether the association between CMI and incident CVD still existed among adults receiving statin therapy, and further researches are needed to validate the association. Finally, despite adjusting for multiple confounders, there might remain residual confounding factors when analyzing two cohort with inherently different genetics, cultures and locations, such as genetic susceptibility, diet, and pollution levels. These may explain some of the differences in the relationship between CMI and incident CVD risks between these two cohorts. Future studies are needed to confirm our findings and explore the underlying mechanism of the differences between two population groups.

5. Conclusions

In the current study, we found that both baseline and total CMI were associated with an increased risk of CVD. Therefore, dynamic monitoring of CMI levels is vital for CVD prevention in clinical practice.

Data availability

Data supporting the results of this study are available from the official websites of China Health and Retirement Longitudinal Study (<http://charls.pku.edu.cn>) and English Longitudinal Study of Ageing (<https://ukdataservice.ac.uk>).

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Ethics approval and consent to participate

The China Health and Retirement Longitudinal Study was approved by the Ethics Review Committee of Peking University. The English Longitudinal Study of Ageing was approved by the Ethics Review Committee of London Multi-Centre Research. Informed consent was obtained from each subject in these two cohorts.

CRedit authorship contribution statement

Jiangtao Li: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiang Wei:** Writing – review & editing, Validation, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ajpc.2025.101046](https://doi.org/10.1016/j.ajpc.2025.101046).

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